12. The density of a material in CGS system of units is 4 g cm^{-3} . In a system of units in which unit of length is 10 cm and unit of mass is 100 g, then the value of density of material will be

a) 0.04 b) 0.4

c) 40 d) 400

13. If the force is proportional to square of velocity, then the dimension of proportionality constant is [JEE-2000]

a) $[MLT^0]$ b) $[MLT^{-1}]$

c) $[ML^{-2}T]$ d) $[ML^{-1}T^0]$

14. The dimension of $\left(\mu_0\varepsilon_0\right)^{-\frac{1}{2}}$ is [Main AIPMT 2011]

(a) length (b) time

(c) velocity (d) force

15. Planck's constant (h), speed of light in vacuum (c) and Newton's gravitational constant (G) are taken as three fundamental constants. Which of the following combinations of these has the dimension of length? [NEET 2016 (phase II)]

(a) $\frac{\sqrt{hG}}{c^{\frac{3}{2}}}$ (b) $\frac{\sqrt{hG}}{c^{\frac{5}{2}}}$

(c) $\sqrt{\frac{hc}{G}}$ (d) $\sqrt{\frac{Gc}{h^{\frac{3}{2}}}}$

Answers:

1) a	2) d	3) d	4) c
5) d	6) c	7) b	8) b
9) a	10) d	11) b	12) c
13) d	14) c	15) a	

II. Short Answer Questions

1. Briefly explain the types of physical quantities.

2. How will you measure the diameter of the Moon using parallax method?

3. Write the rules for determining significant figures.

4. What are the limitations of dimensional analysis?

5. Define precision and accuracy. Explain with one example.

III. Long Answer Questions

1. i) Explain the use of screw gauge and vernier caliper in measuring smaller distances.

ii) Write a note on triangulation method and radar method to measure larger distances.

2. Explain in detail the various types of errors.

3. What do you mean by propagation of errors? Explain the propagation of errors in addition and multiplication.

4. Write short notes on the following.

a) Unit

b) Rounding - off

c) Dimensionless quantities

5. Explain the principle of homogeneity of dimensions. Give example.

IV. Exercises

1. In a submarine equipped with sonar, the time delay between the generation of a pulse and its echo after reflection from an enemy submarine is observed to be 80 s. If the speed of sound in water is 1460 ms^{-1}. What is the distance of enemy submarine? Ans: (58.40 km)

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